PRACTICAL 9(A)

Aim: Create an API endpoint to calculate the factorial of a number and test it using Postman.

Steps / Algorithm

1. Create ASP.NET Core Web API Project

1. Open Visual Studio 2026.
2. Select Create a new project.
3. Choose ASP.NET Core Web API and click Next.

2. Configure Project

4. Project Name: FactorialAPI
5. Select Framework: .NET 8 or the latest available version.
6. Enable OpenAPI Support, if available, and click Create.

3. Create the Controller

7. In Solution Explorer, right-click the Controllers folder.
8. Select Add → Class and name the file MathController.cs.

4. Write the Factorial Logic

9. Replace the contents of MathController.cs with the following code:

```
using Microsoft.AspNetCore.Mvc;

namespace FactorialAPI.Controllers
{
    [ApiController]
    [Route("api/[controller]")]
    public class MathController : ControllerBase
    {
        [HttpGet("factorial/{number}")]
        public IActionResult GetFactorial(int number)
        {
            if (number < 0)
            {
                return BadRequest("Factorial not possible for negative numbers.");
            }

            long factorial = 1;

            for (int i = 1; i <= number; i++)
            {
                factorial *= i;
            }

            return Ok(new
            {
                Number = number,
                Factorial = factorial
            });
        }
```

```
        }
    }
```

Note: This loop calculates factorials for non-negative integers. The long data type can overflow for large inputs; use small test values such as 0 to 20.

5. Run the Project

10. Press F5 or click Run to start the project.
11. Note the HTTPS localhost address shown in the browser or Visual Studio output, for example https://localhost:44390.

6. Test the API in a Browser

12. Open a browser and enter the URL below, replacing the port with your project's actual HTTPS port:

```
https://localhost:44390/api/math/factorial/5
```

13. The response should be JSON similar to:

```
{
  "number": 5,
  "factorial": 120
}
```

7. Test Using Postman

14. Open Postman and create a new HTTP request.
15. Select the GET method.
16. Enter the URL using your project's actual port:

```
https://localhost:44390/api/math/factorial/5
```

17. Click Send. The response should return status 200 OK and the factorial value 120.

8. Test a Negative Number

18. In Postman or the browser, enter the following URL, replacing the port if needed:

```
https://localhost:44390/api/math/factorial/-2
```

19. Send the request. The API should return a 400 Bad Request response with this message:

```
Factorial not possible for negative numbers.
```

Result

The ASP.NET Core Web API endpoint was created successfully to calculate the factorial of a number. The endpoint was tested in a browser and Postman for a valid number and a negative number.